Noisy Boi DIY HOTAS

Cockpit Setup & Accessories Guide



Applies to	Details
Joystick	HFB Joystick with any grip (Felon, Flanker, Raptor, Hornet), right or left hand
Throttle	Twin Engine Throttle (Felon, Raptor)
Collective	Black Shark Collective
Companion documents	Assembly Instructions and Electronics, Firmware & Configuration Guide of each device

1.- Overview

Every device in the ecosystem can be used on the desk, clamped to the edge of the desk or mounted on the chair. This guide covers the accessories that make each setup possible, the joystick extensions and cams, and the throttle detents you can customize.

1.1.- Choosing a setup

Setup	Best for	Accessories
Desktop	Quick setup, shared desks, trying the devices out	Joystick base supports. The throttle sits directly on the desk.
Desk-mounted	**Side stick** (recommended for the joystick): the grip sits higher and is easy to position. The devices hang below the desk edge, leaving the desk free.	Joystick desk mount, throttle desk mount, optional straight extension
Chair-mounted	**Center stick**: the joystick mounts at the center of the chair, low and close to the body. Throttle and collective on the chair armrest move with you.	Joystick, throttle and collective chair mounts
HOSAS	Space sims and dual-stick setups	Two joysticks (one with a mirrored grip), Omnithrottle angled extension, left-hand mini stick mount

1.2.- Accessory index

Accessory	Device	Editable file
Cams (standard, soft center, heavy)	Joystick	—
Straight extension	Joystick	`Estansion.step` (length)
Omnithrottle angled extension	Joystick (HOSAS)	—
Left-hand mini stick mount	Joystick (mirrored grips)	—
Base supports (desktop)	Joystick	—
Desk mount	Joystick	—
Chair mount	Joystick	—
Interchangeable detents	Throttle	`BlankDetent.step` (detent profile)
Desk mount	Throttle	—

Accessory	Device	Editable file
Chair mount	Throttle	—
Chair mount	Collective	—

1.3.- Printing load-bearing parts

Mounts, clamps and extensions carry the full force of your hand on a long lever. Print them stronger than the rest of the parts:

- **Material:** PETG or PLA+. Avoid regular PLA for clamps and arms; it creeps under constant load.
- **Walls:** 4 or more perimeters. Strength comes mainly from the walls, not the infill.
- **Infill:** 40% or more (gyroid or cubic).

2.- Joystick accessories

2.1.- Gimbal cams and springs

The gimbal's dual-spring cams set how the stick feels around the center and how hard it pushes back toward it. Three cam profiles are included:

Cam profile	Feel
Standard	Balanced, general-purpose feel.
Soft center	Light force around the center for fine corrections, building up toward the ends of travel.
Heavy	Higher force through the whole travel. Good with extensions or for a firmer, fighter-like stick.

Springs

Because of the weight of the grip, **use 2 sets of springs per axis**, so the stick returns to center reliably. If it still feels light (for example with an extension), switch to the heavy cams or tighten the dry-clutch dampeners a little.

2.2.- Straight extension

Raises the grip above the base. A longer lever gives finer control over small deflections, as on real fighter sticks and center-mounted setups, and puts the grip at a comfortable height when the base is mounted low.



Straight extension on the joystick base

Item	Details
Default length	100 mm (10 cm)
Editable file	`Estansion.step`
Extra hardware	**One additional pair of 4-pin 360° pogo connectors** for each extension (one at each end)

Choosing the length

- **Maximum recommended length: 15 cm.** The gimbal springs are tuned for the grip alone. With a longer lever your hand gains leverage over them, so the stick feels lighter, centers less firmly and puts more load on the gimbal and the mount.
- If the stick feels too light with the extension, use the heavy cams (section 2.1).
- No recalibration is needed: the sensors are in the gimbal, and the extension doesn't change the deflection angle.

Editing the length

1. Open `Estansion.step` in your CAD software (Fusion, FreeCAD, Onshape, SolidWorks…).
2. Change only the length of the tube section.
3. Do not modify the two ends: they match the base and grip interfaces, the pogo connectors and the screw holes.
4. Export as STL and print.

Assembly follows the same steps and hardware as the main components of the HOTAS ecosystem; the image shows how the parts go together.

2.3.- Omnithrottle angled extension (HOSAS)

The Omnithrottle is an angled extension that offsets the grip from the base and holds it at an angle better suited to HOSAS use than the upright straight position. Combined with a mirrored grip, it turns two HFB joysticks into a HOSAS (Hands On Stick And Stick) setup, the standard layout for space simulators and a popular alternative for helicopters and VTOL aircraft.



Omnithrottle angled extension on the joystick base

Like the straight extension, the Omnithrottle needs **one additional pair of 4-pin 360° pogo connectors**. Assembly follows the same steps and hardware as the main components of the HOTAS ecosystem; the image shows how the parts go together.

Left-hand mini stick mount

All grips can be printed mirrored for the left hand and keep every function. The only part that cannot simply be mirrored is the mount of the analog mini stick, because the module itself is not symmetric. Use the dedicated left-hand mini stick mount instead of the mirrored one.

Thanks

The left-hand mini stick mount was designed and shared by **Kassio L. M. (aka DeadMeat)**.

HOSAS in Windows and games

Two HFB joysticks show up as two separate controllers. Give each one its own PID and USB name (for example "HFB Joystick L" and "HFB Joystick R") as described in the Electronics, Firmware & Configuration Guide, so games never mix up their bindings.

2.4.- Base supports (desktop use)

Four legs that attach to the corners of the joystick base, to use it free-standing on the desk without a mount. The wide footprint keeps the base stable during hard maneuvers.



Joystick base on its desktop supports

- Add rubber feet or anti-slip pads under the leg tips, so the base doesn't slide.
- With long extensions the lever can tip a free-standing base; use the desk or chair mount instead.

2.5.- Desk mount

A C-clamp arm that hangs the joystick from the edge of the desk, so the grip ends up at a natural height for your arm without taking space on the desk. **This is the recommended mount for a side stick:** the grip sits higher than with the chair mount and is easier to position.



Joystick desk mount



Desk-mounted joystick

The mount is assembled with **M3x16 mm** screws.

Before mounting

Recommended desk thickness: **18 mm to 35 mm**.

Don't clamp to glass tops or to thin particle board: the clamping force and the lever can crack or crush them.

Tighten the clamp gradually and check it again after the first session.

2.6.- Chair mount

A plate that attaches the joystick under the center of the seat, as a **center stick** between your knees. The stick moves with you, so your arm rests in the same position every session. It sits lower and closer to your body than the desk mount; for a side stick, use the desk mount (section 2.5).



Joystick chair mount plate



Chair-mounted joystick, side view

Chair compatibility

The plate is fixed to the underside of the seat with **wood screws**. Check that the seat has a solid wooden or plywood base where the screws can bite.

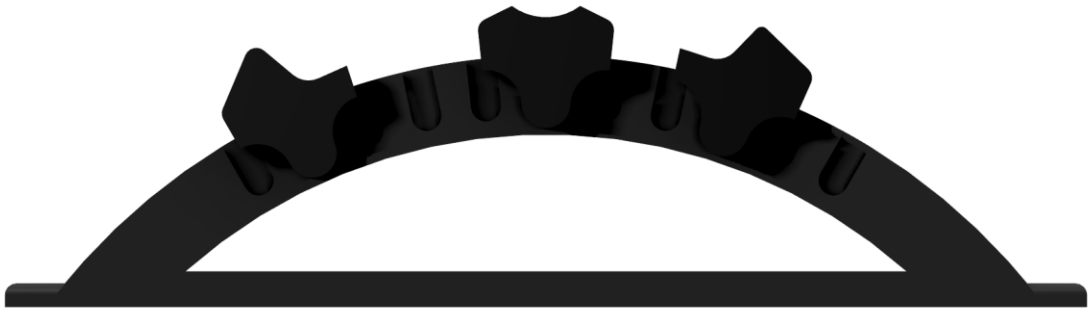
It has not been tested with every chair model or with commercial cockpit mounts.

Leave enough cable slack for the chair to swivel, tilt and change height without pulling the USB cable. A powered USB hub fixed to the chair keeps the cables short.

3.- Throttle accessories

3.1.- Interchangeable detents

The detents give each throttle lever its tactile stops, such as idle cut-off and afterburner. They are screwed to a bracket held by two M2.5x6 mm screws, so you can swap detent profiles to match the aircraft you fly.



Detent bracket with detents

Designing your own detents

1. Open `BlankDetent.step` in your CAD software.
2. Shape the detent profile; keep the mounting base and its screw holes unchanged, so it fits the bracket.
3. Print a test piece first. A steeper ramp gives a firmer detent; a shallower one gives a softer one.

Swapping detents

1. Remove the two M2.5x6 mm screws that hold the detent bracket.
2. Swap the detents on the bracket (M2x5 mm screws).
3. Put the bracket back.
4. If the new profile changes where the lever travel ends, recalibrate the throttle (Tactical MFD Control Hub, CTRL › CAL).

3.2.- Desk mount

Hangs the throttle from the edge of the desk with a C-clamp, at the same height as a desk-mounted joystick.



Throttle desk mount

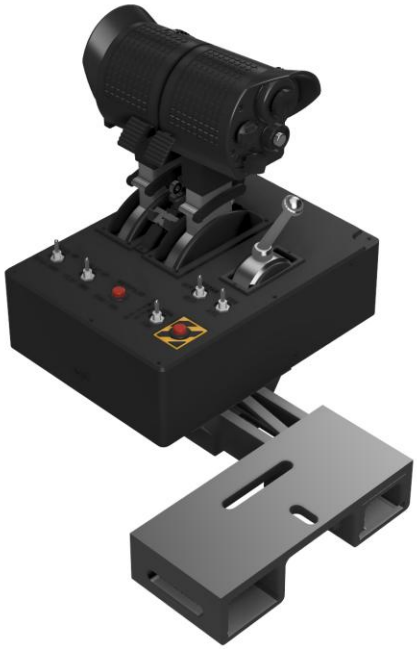


Desk-mounted throttle

The mount is assembled with **M3x16 mm** screws. Recommended desk thickness: **18 mm to 45 mm**. The precautions of section 2.5 also apply.

3.3.- Chair mount

Attaches the throttle to the side of the chair, so it moves with you. It is designed for **standard office chairs with armrests**: the mount bolts into the armrest mounting holes of the chair.



Throttle chair mount



Chair-mounted throttle

Assembly follows the same steps and hardware as the main components of the HOTAS ecosystem; the image shows how the parts go together. Mount the throttle at the same height as the joystick, so both arms rest at the same angle. Compatibility with every chair model or with commercial cockpit mounts has not been tested. Leave enough cable slack for the chair to swivel and change height.

4.- Collective accessories

4.1.- Chair mount

In a helicopter the collective sits on the left side of the seat and moves up and down with the pilot's forearm. The chair mount reproduces that position: the lever pivot ends up next to your hip, and the grip under your left hand. Like the throttle chair mount, it is designed for **standard office chairs with armrests** and bolts into the armrest mounting holes of the chair.



Collective chair mount



Chair-mounted collective

5.- Ergonomics tips

Item	Recommendation
Joystick height	Forearm roughly horizontal and elbow close to 90° with the hand on the grip. Use the straight extension to raise the grip, or lower the mount.
Side stick vs. center stick	For a side stick, use the joystick desk mount: it puts the grip higher and is easier to position. The joystick chair mount is meant for a center stick. In both cases, rest the forearm so the wrist, not the shoulder, moves the stick.
Throttle position	Same height as the joystick, close enough to reach every switch without leaning.
Collective	Pivot next to the hip; the forearm should rest along the lever at mid travel.
Pedals	Knees slightly bent at full deflection. Fix the pedals to a board or the floor so they don't slide.
Cables	Use a powered USB hub. Route cables away from moving parts and leave slack for every axis and for the chair.

For questions contact me at: the.noisy.boi@gmail.com

ALL DONE – HAPPY FLYING!